

HUJI Applicative Research Report — Exact & Social Sciences — Week 33, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 33, 2026 · Generated August 10, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Hebrew University's latest research showcases exciting commercial potential across both deep tech and applied solutions. One groundbreaking discovery demonstrates how chiral perovskites can enable contactless optical sensing and advanced clean energy conversion. Complementing this, an innovative study reveals a method to sustain healthier food choices on online platforms, offering a robust solution for the evolving FoodTech landscape.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Igal Levine** — Materials, Clean Energy, Quantum
- **Roni Lotan** — FoodTech, Software/AI, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Contactless Detection of Giant Helicity-Dependent Photovoltage in Chiral 2D Perovskites. 25/50

"Chiral perovskites enable contactless optical sensing and breakthrough clean energy conversion."

Main researcher: Igal Levine

Institute of Chemistry and the Center for Nanoscience and Nanotechnology, the Hebrew University of Jerusalem, Jerusalem, Israel.
Published in Small (Weinheim an der Bergstrasse, Germany) · 2026 Aug 8

ABOUT THIS RESEARCH

Researchers have developed a method to detect high-voltage electrical signals in chiral 2D perovskite materials using light with specific rotational polarity, all without physical contact. This discovery demonstrates a highly efficient way to convert circularly polarized light into electricity using asymmetric crystal structures.

COMMERCIAL ANGLE

Development of ultra-sensitive, non-contact optical sensors and next-generation spintronic devices for secure communications and advanced light-harvesting energy cells.

COMMERCIALISATION METRICS

Novelty

7/10

The integration of contactless detection with giant helicity-dependent photovoltage in 2D perovskites represents a significant technical leap over traditional contact-based measurements, though it builds on established chiral semiconductor physics.

Commercial Potential

4/10

While the discovery of giant photovoltage in chiral 2D perovskites has long-term potential for optoelectronics and sensing, the lack of a defined application or device architecture makes the immediate commercial path speculative.

Market Size

6/10

The discovery of giant photovoltage in chiral 2D perovskites suggests broad potential for next-generation optoelectronics, sensors, and quantum computing, though specific commercial applications remain early-stage.

Tech Readiness

2/10

The research focuses on fundamental physical phenomena (helicity-dependent photovoltage) in a novel material class, representing basic laboratory-stage discovery rather than an applied device or commercial prototype.

IP Strength

6/10

The innovation involves a specific material class (chiral 2D perovskites) and a novel detection method, which is patentable; however, defensibility is moderate as it relies on material composition and physical effects that may be bypassed by alternative chiral semiconductors.

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2. Beyond one-shot nudges: the effects of repeated interventions for healthier online meal ordering.22/50

"New behavioral AI prevents "nudge decay" in food apps, driving healthier long-term choices."

Main researcher: Roni Lotan

School of Public Policy, The Hebrew University of Jerusalem, Jerusalem, Israel.; For a Change, Research Lab, Tel Aviv, Israel.
Published · 2026-05

ABOUT THIS RESEARCH

The study investigates how different digital nudges, specifically real-time feedback and assortment changes, influence healthy food choices in online ordering platforms over multiple days. It finds that combining structural changes with moment-of-choice feedback prevents the common 'decay effect' where users revert to old habits, maintaining higher health scores over time.

COMMERCIAL ANGLE

Developing specialized UI/UX behavioral optimization software for food delivery and grocery platforms to drive long-term consumer adherence to nutritional guidelines.

COMMERCIALISATION METRICS

Novelty4/10

The research applies well-established behavioral economics principles (nudging and assortment) to a known domain, offering incremental insight into temporal decay rather than introducing a novel mechanism or disruptive technology. While the longitudinal focus adds value to digital health strategy, it does not represent a fundamental scientific breakthrough or a new platform-enabling methodology.

Commercial Potential6/10

The research identifies a specific, implementable UX strategy (combining assortment and feedback) that has direct B2B licensing potential for food delivery platforms looking to improve ESG metrics or user health outcomes. However, it lacks a proprietary technical moat or deep-tech component, functioning more as an optimization playbook for existing digital interfaces.

Market Size7/10

The target market is massive, encompassing the global digital food delivery and e-commerce sectors where even marginal improvements in consumer choice behavior drive significant commercial value. However, the application is a feature-level optimization rather than a standalone platform, limiting its direct capture of the total addressable market.

Tech Readiness3/10

The technology is at a fundamental/proof-of-concept stage, utilizing a mock platform and hypothetical orders rather than a live commercial environment. While the behavioral insights are promising, the study lacks integration into real-world transactional systems or large-scale user environments.

IP Strength2/10

The innovation relies on behavioral economics principles (nudging, assortment, and feedback) which are widely documented in academic literature and are generally considered non-patentable psychological interventions. There is no evidence of a proprietary algorithm, novel software architecture, or unique hardware that would provide a defensible technical moat.

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